

Comprehensive Analytics Framework for Event-Driven Urban Traffic Congestion: Predictive Modeling and Resource Optimization

1. Introduction to the Dynamics of Event-Driven Congestion

Urban traffic systems represent one of the most complex, non-linear, and dynamic environments modeled by modern data science. Within these metropolitan networks, congestion is generally categorized into recurrent and non-recurrent paradigms. Recurrent congestion follows predictable temporal cycles dictated by daily commuting behaviors and localized infrastructure constraints. Conversely, non-recurrent or event-driven congestion fundamentally disrupts these equilibrium states, injecting stochastic shocks into the transportation graph.¹ These operational shocks originate from planned events, such as political rallies, religious festivals, sports matches, and scheduled construction activities, as well as unplanned incidents, including vehicular breakdowns, sudden infrastructural failures, and multi-vehicle accidents. The operational challenge posed by these events is immense, as localized traffic breakdowns rapidly propagate through arterial corridors, creating compounding gridlock that overwhelms traditional management strategies.³

Contemporary traffic management systems face three acute operational deficiencies when confronting event-driven congestion. First, the impact of an event is rarely quantified in advance; managers lack deterministic models to forecast the exact spatial reach and temporal duration of a disruption before the shockwave paralyzes the network. Second, the deployment of critical mitigation resources—ranging from traffic police manpower to physical barricading and vehicular diversion routing—is almost exclusively experience-driven. Command centers rely on the intuition of veteran officers rather than algorithmic optimization, leading to inefficient resource allocation where secondary roads are inadvertently flooded with diverted traffic.⁵ Third, there is a pervasive absence of systematic post-event learning. Once an incident is resolved, the operational data is rarely curated and fed back into a machine learning pipeline to update historical baselines, leaving the system perpetually reactive rather than proactive.⁷

This exhaustive research report establishes a robust, data-driven framework designed to resolve these deficiencies. By dissecting high-resolution incident data, synthesizing predictive feature engineering techniques, and benchmarking against cutting-edge methodologies developed in global machine learning competitions—including those hosted by Kaggle, NeurIPS, Harvard University, and Stanford University—this analysis constructs a comprehensive mathematical and operational blueprint. The objective is to demonstrate how historical and real-time data can be leveraged to not only forecast event-related traffic impacts but also to

algorithmically recommend optimal manpower allocation, barricading strategies, and diversion plans, ultimately establishing an intelligent, continuously learning urban mobility ecosystem.

2. Anatomical Exploration of the Traffic Incident

Dataset

The foundation of any predictive traffic management framework is the underlying dataset. A rigorous exploration of the provided anonymized event data reveals a highly structured, multi-dimensional repository that captures the geospatial, temporal, and categorical attributes of urban traffic disruptions across various metropolitan zones.³ Understanding the schema, identifying latent variables, and recognizing the intrinsic biases of this data are essential prerequisites for constructing advanced machine learning features.

The dataset provides a macroscopic view of incident generation across a major urban environment. It encapsulates a wide spectrum of operational realities, distinguishing between planned public events, such as cricket matches at central stadiums, and highly stochastic unplanned events, such as sudden vehicular mechanical failures, water logging, and tree falls.³

To systematically approach this data, the variables must be grouped into analytical domains.

Data Domain	Representative Features	Analytical Value and Operational Implications
Geospatial Coordinates	latitude, longitude, endlatitude, endlongitude	Enables exact point-mapping of incidents onto a graph-based road network. The inclusion of end coordinates allows for the measurement of incident spatial footprints (e.g., a procession or a moving construction zone).
Categorical Localization	address, zone, police_station, corridor, junction	Facilitates the clustering of incidents by administrative boundaries and infrastructure types. Categorizing an event as occurring on a high-capacity corridor (e.g., "Tumkur Road") versus a residential street dictates the severity of the expected congestion

		propagation.
Temporal Lifecycle	start_datetime, resolved_datetime, closed_datetime, created_date	Critical for calculating the Incident Clearance Time (ICT) and the reporting latency. The delta between incident creation and resolution forms the primary target variable for survival analysis models.
Incident Typology	event_type, event_cause, requires_road_closure, priority	Differentiates between predictable shocks (planned) and sudden disruptions (unplanned). The road closure boolean is the primary trigger for initiating diversion routing algorithms.
Vehicular Metadata	veh_type, veh_no, cargo_material, reason_breakdown, age_of_truck	Provides essential parameters for estimating recovery complexity. Heavy vehicles carrying dense cargo experience vastly different recovery timelines compared to standard passenger vehicles.
Unstructured Context	description, comment	Contains free-text, often multilingual human observations that provide nuanced severity indicators not captured by standardized drop-down fields.
Response Mechanics	assigned_to_police_id, resolved_by_id, closed_by_id	Tracks the human resource chain of custody, enabling the evaluation of personnel efficiency and the

		optimization of future manpower allocation.
--	--	---

A detailed examination of the dataset highlights several critical operational realities. For instance, the data delineates heavy reliance on categorical event causes such as vehicle_breakdown, water_logging, tree_fall, and accident.³ Each of these categories exhibits highly distinct temporal recovery profiles. A tree fall on a secondary road, such as the incident recorded on Sankey Road (FKID000003), often necessitates coordination with municipal forestry or clearance departments, leading to prolonged resolution times extending over several hours or days.³ Conversely, minor vehicular breakdowns, such as a private car stalling due to a starting problem on an outer ring road, are typically resolved within an hour.³ The dataset also captures the severe impact of commercial and public transit vehicle failures. Entries detailing breakdowns of bmtc_bus or heavy_vehicle assets frequently cite mechanical failures such as "clutch plate got broken," "gear box problem," or "axle cut".³ Furthermore, the inclusion of variables like cargo_material and age_of_truck adds a profound layer of predictive utility. An incident involving a 15-year-old heavy truck carrying construction materials (e.g., hardware or cement) that suffers an axle failure on a major flyover represents a worst-case scenario for recovery logistics.³ Extracting these nuances is vital because the physical effort, specialized machinery (e.g., heavy-duty towing cranes), and time required to clear such an obstruction directly dictate the duration of the capacity reduction on that specific road segment.

Equally important is the unstructured data contained within the description and comment fields. These fields often feature code-switched text combining English and regional languages (e.g., Kannada), such as "ಉರ್ವಶಿ ಜಂಕ್ಷನ್ ನಲ್ಲಿ ಬಳಚರಂಡಿ ಚೇಂಬರ್ ಗೆ ಹೊಸದಾಗಿ ಸಿಮೆಂಟ್ ಹಾಕಿದ್ದು ಟ್ರಾಫಿಕ್ ಮೂವ್ಮೆಂಟ್ ಸ್ವಲ್ಪ ನಿಧಾನಗತಿಯಲ್ಲಿ ಇರುತ್ತದೆ ಸರ್" (New cement laid for drainage chamber at Urvashi junction, traffic movement will be slightly slow).³ Such entries contain highly specific, localized intelligence regarding infrastructure conditions that structured categorical fields completely fail to capture. Mining this text allows the system to differentiate between a total blockage and a scenario where traffic is merely experiencing "slow movement" due to a partial obstruction.

Finally, the presence of planned events provides an opportunity for deterministic forecasting. Events labeled public_event or procession typically have predefined start and end times, geographic boundaries, and known expected attendance sizes. For instance, a cricket match at a major stadium (FKID000008) generates massive, predictable surges in traffic volume both before and after the event.³ By analyzing historical data from similar past events, predictive models can anticipate the exact spatial and temporal dimensions of the traffic surge, allowing for the preemptive deployment of traffic management resources before the first bottleneck even forms.

3. Advanced Feature Engineering for Traffic Impact

Quantification

To successfully transition from raw observational data to a sophisticated predictive architecture capable of quantifying event impact in advance, the raw dataset must undergo rigorous feature engineering. Machine learning models require mathematically represented inputs to discover complex, non-linear relationships. Drawing upon advanced data science methodologies utilized in the transportation domain, several distinct feature spaces must be constructed from the provided dataset.⁹

3.1. Cyclical Temporal Encoding

Urban traffic patterns are deeply periodic, governed by the rhythmic nature of human activity, including daily commuting cycles, school hours, and weekly variations in commercial logistics.¹ Traditional representations of time (e.g., hour of the day as an integer from 0 to 23) are problematic for machine learning algorithms because they fail to convey the continuous, cyclical nature of time. For example, an algorithm interpreting standard integers views 23:00 and 00:00 as being far apart, despite them being chronologically adjacent. To resolve this, temporal variables must be transformed using sine and cosine functions. By applying these trigonometric transformations, the exact time of an incident's inception is mapped onto a continuous circle.

$$Time_{sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad Time_{cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right)$$

$$Day_{sin} = \sin\left(\frac{2\pi \cdot \text{day_of_week}}{7}\right), \quad Day_{cos} = \cos\left(\frac{2\pi \cdot \text{day_of_week}}{7}\right)$$

In addition to continuous cyclical encoding, discrete boolean flags must be engineered to capture macroscopic temporal states. Variables such as `is_weekend`, `is_public_holiday`, and `is_peak_hour` provide essential context.¹⁰ An identical vehicular breakdown will yield vastly different congestion impacts if it occurs at 03:00 on a Sunday versus 08:30 on a Monday morning.

3.2. Graph-Based Spatial Representations

Traffic congestion does not exist in isolation; it behaves as a fluid dynamic that propagates through a highly connected spatial network.¹² Utilizing isolated latitude and longitude points is insufficient for predicting network-wide impact. The spatial coordinates from the dataset must be projected onto a directed graph $\mathcal{G} = (V, E)$, where V represents intersections and E represents the connecting road segments.

Once mapped to a graph, powerful spatial features can be derived. The distance from the incident location to critical infrastructure nodes—such as major transit hubs, hospitals, or primary highway interchanges—must be calculated. Furthermore, network centrality metrics,

such as Betweenness Centrality or Eigenvector Centrality, should be assigned to the corridor where the incident occurs. An event taking place on a high-centrality arterial route, such as the Outer Ring Road (ORR), possesses a significantly higher capacity to paralyze surrounding networks than an event on a low-centrality peripheral road.³ Additionally, integrating spatial flow inputs from upstream and downstream loop detectors or traffic cameras provides critical context regarding the pre-existing load on the network at the time of the incident.¹⁴

3.3. Incident Recovery Complexity Index

The duration of a lane closure is the primary determinant of event-driven congestion severity. To predict this duration, a composite `Recovery_Complexity_Score` must be synthesized by fusing multiple vehicular and situational parameters.³ This index serves as a proxy for the logistical difficulty of clearing the obstruction.

The algorithm for this index must heavily weight attributes such as `veh_type`, `age_of_truck`, and `cargo_material`. For example, a dataset entry indicating the breakdown of a heavy commercial vehicle transporting raw steel, complicated further by a severe mechanical failure like a sheared axle, receives a maximum complexity score.³ This high score mathematically signals to the subsequent predictive models that specialized recovery equipment will be required, inter-agency coordination may be necessary, and the time-to-clearance will be significantly extended.

3.4. Multilingual Natural Language Processing (NLP)

A massive reservoir of latent predictive power resides within the unstructured text of the description and comment fields. Relying solely on structured categorical inputs overlooks critical ground-truth realities reported by on-site personnel or citizens.³ To harness this information, advanced Natural Language Processing (NLP) pipelines leveraging Large Language Models (LLMs) and Transformer architectures are required.¹

The NLP pipeline must address the multilingual and code-switched nature of the data, seamlessly processing both English and regional dialects. By passing the text through a customized embedding model (such as a fine-tuned multilingual BERT), unstructured observations are converted into dense numerical vectors that capture semantic meaning. This allows the system to automatically flag high-urgency contextual clues—such as "fuel leakage," "chain accident," or "water logging with drainage leakage"—and translate them into a quantitative `Urgency_Score` that directly informs the predictive models regarding the immediate threat to public safety and network fluidity.³

4. Methodological Benchmarking: Insights from Global Competitions

The theoretical foundation for solving complex urban mobility challenges has been dramatically advanced by global machine learning competitions, academic hackathons, and corporate research initiatives. By analyzing the winning architectures and methodologies from these highly competitive arenas, we can validate and refine the predictive strategies necessary for addressing event-driven congestion.

4.1. The Kaggle Geotab Intersection Congestion Challenge

Hosted on the Kaggle platform, the "BigQuery-Geotab Intersection Congestion" competition tasked data scientists globally with predicting traffic congestion metrics across four major United States cities.¹⁷ The specific objective was to forecast the 20th, 50th, and 80th percentiles for total wait times and stopping distances at complex intersections, utilizing a massive dataset of aggregated commercial vehicle telematics.¹⁸

An analysis of the winning solutions from this competition yields several critical insights applicable to our framework. Top-performing models largely abandoned attempts to build a single, monolithic global model, finding them too brittle to account for extreme localized variance. Instead, competitors achieved superior results by building highly localized models or by utilizing gradient boosting frameworks, specifically LightGBM and XGBoost, augmented with exhaustive spatial-temporal feature aggregations.¹⁷

Furthermore, the competition highlighted the absolute necessity of advanced categorical feature encoding. Winning participants utilized target encoding strategies for intersection IDs to capture historical congestion baselines.²² More importantly, the requirement to predict multiple quantiles (20th, 50th, and 80th percentiles) rather than a single deterministic mean is a crucial methodological takeaway. In the context of our event-driven traffic model, predicting a probabilistic range for an incident's clearance time provides traffic commanders with a nuanced risk profile. Knowing that a breakdown has a 50% chance of clearing in 30 minutes, but an 80% chance of extending to 90 minutes, allows for vastly superior, risk-adjusted resource deployment.

4.2. NeurIPS Traffic4cast and the Paradigm of Image Segmentation

The Institute for Advanced Research in Artificial Intelligence (IARAI) fundamentally altered the landscape of traffic prediction through the Traffic4cast competitions hosted at the Neural Information Processing Systems (NeurIPS) conference.²³ Traditionally, traffic forecasting was treated purely as a tabular time-series problem. Traffic4cast redefined the challenge by transforming it into a spatial-temporal *movie completion task*.²⁴ Participants were provided with vast amounts of GPS probe data mapped into multi-channel image tensors, where pixels represented the road network and color channels indicated congestion classes (red, yellow, green) over time.²⁶

The architecture that consistently dominated this competition was the **U-Net**—a type of Convolutional Neural Network (CNN) featuring an encoder-decoder structure with skip connections, originally designed for precise biomedical image segmentation.¹² The success of U-Nets demonstrated their unparalleled ability to extract complex spatial-temporal features from real-world geospatial processes.

A particularly surprising insight from the Traffic4cast results was the realization that providing the U-Net with just one hour of historical dynamic traffic data, combined with a static map of the road graph, was entirely sufficient to accurately predict traffic states up to 15 minutes into the future.¹² The winning solutions proved so robust that they successfully managed severe domain shifts, maintaining accuracy when tested on entirely unseen cities or when predicting

traffic patterns in a post-COVID environment based on pre-COVID training data.¹² Translating this to our operational framework implies that by converting our geospatial point data and real-time flow metrics into a spatial-temporal grid, we can utilize U-Net architectures to visually forecast the ripple effect and geographical spread of a congestion shockwave originating from an unplanned event.

4.3. Stanford Open Policing and Fairness in Resource Allocation

Academic research from Stanford University, particularly the Stanford Open Policing Project, provides a masterclass in the application of machine learning to vast, real-world law enforcement datasets.²⁸ This initiative compiled and analyzed over 200 million traffic stop records across the United States to investigate patterns in policing, resource deployment, and algorithmic fairness.¹³

The project utilized a blend of unsupervised and supervised learning techniques. Unsupervised clustering methods, including K-means and Hierarchical Clustering, were deployed to categorize geographic regions and demographic profiles based on stop outcomes and operational variables.²⁹ Supervised models, such as Logistic Regression and Naive Bayes, were subsequently trained to predict stop outcomes while simultaneously being constrained by mathematical fairness metrics to prevent algorithmic bias.³⁰

The application of these techniques is highly relevant to optimizing traffic manpower. Just as the Stanford project clustered geographic zones based on policing activity, our framework can deploy K-means clustering on the historical event data to identify persistent spatial "hotspots" for specific incident types (e.g., identifying localized zones highly prone to water logging during specific weather conditions).³ This spatial clustering directly informs the strategic, preemptive positioning of traffic police and emergency recovery vehicles, reducing travel times to critical incidents. Furthermore, at hackathons like Stanford TreeHacks, developers have demonstrated the efficacy of Graph Neural Networks (GNNs) combined with attention mechanisms for capturing the spatial relationships between road segments, proving highly effective for low-latency edge computing applications.¹¹

4.4. Harvard Innovations in Policy and Real-Time Control

Harvard University's research ecosystem has contributed significantly to the intersection of machine learning, public policy, and urban infrastructure. A prominent example is the development of the Surtrac system by Rapid-Flow Technologies, a spin-off born from research at Carnegie Mellon and Harvard.³³ Surtrac addresses traffic signal management by moving away from centralized, macroscopic control and instead utilizing machine learning and edge computing directly at the intersection level. By utilizing computer vision to analyze real-time traffic volume and employing decentralized algorithms to optimize signal timing dynamically, the system achieved a 26% reduction in travel times and a 41% reduction in intersection wait times during pilot deployments.³³

Harvard's approach emphasizes that machine learning provides the most value when dealing with interconnected variables changing so rapidly that traditional "automated without learning" systems fail.³³ Furthermore, Harvard's Kennedy School has championed the integration of

crowdsourced data into civic operations.³⁴ By partnering with platforms like Waze, city governments have successfully leveraged citizen-generated data to identify congestion hotspots and evaluate policy changes without the massive capital expenditure of deploying physical sensor networks.³⁴ Additionally, policy-focused "hackathons" at Harvard have been instrumental in preparing municipal leaders for disruptive technologies, such as the introduction of autonomous vehicles or the integration of Transportation Network Companies (TNCs) with public transit.³⁶ These initiatives underscore that predictive algorithms must be deeply integrated with civic policy and human-in-the-loop operational protocols to achieve sustainable impact.

5. Predictive Modeling for Impact Quantification

The translation of engineered features and benchmarked insights into a functional operational tool requires a sophisticated, multi-tiered machine learning architecture. The overarching goal stated in the problem formulation is to quantify the event impact in advance. This necessitates predicting two highly distinct but interrelated metrics: the temporal duration of the incident itself, and the spatial propagation of the resulting traffic congestion.¹

5.1. Modeling Incident Clearance Time (ICT)

The Incident Clearance Time (ICT)—the exact duration required to remove an obstruction, clear a breakdown, or safely resolve an accident—is the primary variable dictating the severity of capacity reduction on a road segment. Predicting ICT is mathematically challenging due to the right-censored nature of ongoing events; an algorithm must predict the total duration of an event while the event is still actively unfolding.³⁹

Hazard-Based Survival Analysis: To address the temporal dynamics of ongoing incidents, survival analysis, specifically hazard-based duration modeling, is highly effective.³⁹ Let T be a continuous random variable representing the incident duration. The survival function $S(t)$ expresses the probability that the incident continues beyond time t :

$$S(t) = P(T > t) = \int_t^{\infty} f(u) du$$

where $f(u)$ represents the probability density function of the clearance times. The hazard function $h(t)$, which is critical for real-time operations, represents the instantaneous rate of clearance at time t , given that the incident has already lasted up to time t :

$$h(t) = \frac{f(t)}{S(t)}$$

By utilizing Accelerated Failure Time (AFT) models or Cox Proportional Hazards models, the traffic management system can continuously and dynamically update the predicted remaining duration of an event as minutes tick by, providing command centers with a constantly evolving risk assessment.³⁹

Ensemble Tree Architectures: While survival models handle the continuous nature of time effectively, ensemble tree architectures—specifically Extreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM)—have consistently demonstrated superior predictive accuracy for immediate regression tasks involving complex, tabular traffic data.⁸ By ingesting the engineered features (e.g., Recovery_Complexity, Time_sin, Graph_Centrality), the

XGBoost algorithm operates by minimizing a regularized objective function at each iteration t :

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t)$$

where l is a differentiable convex loss function measuring the difference between the true ICT y_i and the prediction $\hat{y}_i$, and $\Omega(f_t)$ is the penalty term for model complexity to prevent overfitting:

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

Empirical studies indicate that XGBoost yields superior Mean Absolute Percentage Error (MAPE) when predicting the clearance times of short and medium-duration crashes.⁸ By utilizing SHAP (SHapley Additive exPlanations) values alongside the XGBoost output, traffic commanders can see exactly which features (e.g., the presence of a heavy truck or a specific geographic node) are driving the predicted delay, enhancing the interpretability and trustworthiness of the model.⁸

5.2. Forecasting Spatial Congestion Propagation

Once the ICT is accurately predicted, the system must forecast how the resulting reduction in road capacity will impact the surrounding network. Traditional univariate time-series models, such as ARIMA, are insufficient for this task as they fail to capture the complex spatial dependencies inherent in gridlocked traffic.⁹

Instead, the framework relies on Spatio-Temporal Graph Convolutional Networks (STGCN).¹⁵

The urban road network is formalized as a graph $\mathcal{G} = (V, E, A)$, where V represents the set of intersections, E represents the connecting road segments, and $A \in \mathbb{R}^{N \times N}$ is the

adjacency matrix defining the connectivity. The graph convolution operation at a given layer l allows the model to capture the spatial influence of neighboring nodes:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right)$$

where $\tilde{A} = A + I_N$ (representing the adjacency matrix with added self-loops), $\tilde{D}$ is the degree matrix, $H^{(l)}$ denotes the hidden representations of the traffic state, and $W^{(l)}$ is the layer-specific learnable weight matrix.

The FUSE-Traffic Integration: While GNNs excel at modeling periodic traffic flow, they struggle to adapt to the sudden, chaotic dynamics introduced by non-periodic events. To bridge this gap, the predictive architecture incorporates the FUSE-Traffic framework.¹ This state-of-the-art methodology synergizes the dynamic understanding capabilities of Large Language Models (LLMs) with the spatial-temporal prowess of GNNs.

In this architecture, the LLM processes the unstructured incident reports and contextual metadata to extract rich event semantics. A cross-attention multimodal fusion mechanism is then employed to integrate these semantic embeddings directly with the numerical traffic flow features processed by the STGCN.¹ This multimodal approach enables the model to dynamically perceive and adapt to event-triggered pattern changes—such as understanding that a "procession" will move slowly across multiple nodes over several hours—yielding robust predictive accuracy precisely when traditional traffic patterns are most severely disrupted.¹

6. Algorithmic Optimization of Resource Deployment

Predicting the duration and spread of congestion is an analytical achievement, but solving the operational challenge requires converting those predictions into prescriptive actions. The problem statement explicitly highlights that current resource deployment is experience-driven. To replace human intuition with mathematical precision, the framework utilizes operations research techniques to optimize the allocation of manpower, the placement of barricades, and the execution of diversion plans.⁴⁵

6.1. Dynamic Manpower and Fleet Allocation

Traffic police personnel, rapid response teams, and specialized recovery vehicles (e.g., heavy-duty cranes) are strictly finite resources. During periods of high urban stress, multiple unplanned events may occur simultaneously. Assigning the correct personnel (assigned_to_police_id) to the most critical incidents requires rigorous mathematical optimization.³

The allocation problem is formulated as a Mixed-Integer Linear Programming (MILP) model.⁴⁹

Let I be the set of active traffic incidents, and P be the set of available response units. The primary decision variable, x_{ij} , is a binary integer equal to 1 if response unit j is assigned to

incident i , and 0 otherwise.

Objective Function: The goal is to minimize the aggregate response latency and the subsequent clearance time across the entire network.

$$\min \sum_{i \in I} \sum_{j \in P} (t_{ij}^{travel} + t_i^{clearance}) \cdot x_{ij} \cdot w_i$$

Here, t_{ij}^{travel} represents the dynamic travel time for unit j to reach incident i (updated in real-time based on the STGCN traffic flow predictions), $t_i^{clearance}$ is the machine-learning predicted ICT for that specific event, and w_i is a programmatic priority weight assigned to the incident. For instance, an incident with priority = High occurring on a major arterial corridor is assigned a significantly higher weight than a minor breakdown on a residential street.³

Operational Constraints:

The optimization is bound by strict operational realities:

1. $\sum_{j \in P} x_{ij} \geq 1 \quad \forall i \in I$ (Ensures every registered incident receives at least one responding unit).
2. $\sum_{i \in I} x_{ij} \leq 1 \quad \forall j \in P$ (Ensures a single unit is not simultaneously assigned to multiple disparate incidents).
3. **Skill and Equipment Matching:** The model enforces hard constraints based on equipment needs. A heavy commercial vehicle breakdown mathematically requires a heavy towing unit; the MILP algorithm prevents the dispatch of a standard patrol car to such an event, thereby eliminating wasted response cycles.⁴⁷

By solving this MILP continuously in real-time, the command center transitions from relying on dispatcher intuition to algorithmically ensuring that the closest, most appropriately equipped officers are directed to the nodes where they can achieve the maximum reduction in network delay.⁵¹

6.2. Optimization of Barricading and Diversion Routing

When an event mandates the partial or complete closure of a road segment (e.g., requires_road_closure = TRUE), incoming traffic must be physically halted via barricading and redirected.³ Experience-driven diversion often leads to catastrophic failure, as manually routing thousands of vehicles onto secondary roads that lack the structural capacity or signal timing to handle the surge simply shifts the bottleneck to a new location.⁵

To recommend mathematically sound barricading and diversion plans, the framework employs a **Bi-level Optimization Model** based on transportation network theory.⁵

- **The Upper Level (System Optimum):** Represents the objectives of the traffic management authority. The goal is to minimize the total travel time across the entire affected sub-network by strategically placing physical barricades and utilizing Variable

Message Signs (VMS) to suggest alternative routes.

- **The Lower Level (User Equilibrium):** Represents the behavior of individual drivers. According to Wardrop's First Principle, traffic arranges itself in a state of User Equilibrium (UE) where no individual driver can unilaterally reduce their travel time by switching to an alternate route.

The lower-level UE problem is expressed mathematically as a convex optimization problem:

$$\min_{\mathbf{v}} \sum_{a \in A} \int_0^{v_a} t_a(x) dx$$

Subject to flow conservation and non-negativity constraints:

$$\sum_{k \in K_{rs}} f_k^{rs} = q_{rs} \quad \forall r, s$$

$$v_a = \sum_{r,s} \sum_{k \in K_{rs}} f_k^{rs} \delta_{a,k}^{rs} \quad \forall a$$

where v_a denotes the traffic volume on link a , $t_a(x)$ is the link performance function defining travel time as a function of volume, q_{rs} is the total travel demand between origin r and destination s , and f_k^{rs} is the flow on specific path k .

By feeding the machine-learning predicted capacity reductions into this bi-level model, the algorithm computationally simulates thousands of routing permutations.⁵⁵ It identifies the precise upstream intersections where barricades should be physically deployed to effectively cap the flow into the compromised zone. Simultaneously, it calculates a diversion routing scheme that disperses the rerouted traffic across multiple capable secondary arteries, approximating the System Optimum and preventing the catastrophic overload of any single detour.⁶

7. Architecting a Post-Event Learning System

The final operational challenge identified in the problem statement is the lack of a post-event learning system. A predictive model that is trained once and deployed statically will inevitably degrade as the underlying urban dynamics evolve—a phenomenon known as "concept drift".⁵⁷ Changes in municipal infrastructure, shifts in residential density, or macroeconomic trends continuously alter how traffic behaves.

To combat concept drift and ensure the framework remains relentlessly accurate, an online learning architecture must be implemented.⁵⁸ Unlike static batch training, online learning methodologies continuously update the model weights sequentially as new streams of event data arrive and are resolved.

By tracking statistical divergence metrics—such as the Hellinger distance—between the historical training distribution and the live data stream, the system can automatically detect when concept drift is occurring.⁶⁰ Upon detection, the system triggers algorithms like the Accuracy Updated Ensemble (AUE) or Dynamic Weighted Majority (DWM) to adaptively re-weight the predictive models, heavily favoring recent data patterns over obsolete historical trends.⁶¹

This continuous learning loop is operationalized through platforms like Bengaluru's ASTraM.⁷ As the system ingests data from ANPR cameras, crowdsourced citizen reports, and AI-driven smart cameras (which detect the vast majority of traffic violations autonomously), every resolved incident acts as a new training label.⁷ When an event concludes, the system autonomously calculates the variance between the machine-learning predicted ICT and the actual, ground-truth clearance time.⁴² This error gradient is fed back into the XGBoost and STGCN models via backpropagation, ensuring that the system's capacity to predict the impact of future events is perpetually refined and optimized.

8. Conclusion

The mitigation of event-driven urban traffic congestion is a multifaceted operational challenge that exposes the critical limitations of reactive, experience-driven management. Transitioning to a proactive, highly resilient transit ecosystem requires the abandonment of intuition in favor of quantified, algorithmic precision.

As demonstrated through the rigorous analysis of high-resolution incident data and validated by the winning architectures of global machine learning competitions—from the spatial-temporal segmentation of NeurIPS Traffic4cast to the localized intersection modeling of Kaggle's Geotab challenge—the technological capacity to accurately forecast the impact of sudden urban shocks is fully realizable. By engineering complex temporal, spatial, and semantic features, and processing them through advanced architectures like XGBoost and LLM-fused Graph Neural Networks, traffic authorities can accurately predict both the clearance time of an incident and the geographical propagation of the resulting gridlock.

Crucially, these predictive insights must be weaponized through operations research. By deploying Mixed-Integer Linear Programming to dynamically allocate finite police manpower and utilizing Bi-level User Equilibrium models to mathematically optimize barricading and diversion routing, command centers can enforce mitigation strategies that actively minimize network-wide delays. Finally, by wrapping these algorithms within an online learning framework that continuously updates against concept drift, metropolitan authorities ensure that their intelligent transportation systems evolve in lockstep with the city itself, fostering a sustainable, data-driven urban mobility infrastructure.

Works cited

1. Fusion of Unstructured and Structured Data for Event-aware Traffic Forecasting - arXiv, accessed June 16, 2026, <https://arxiv.org/html/2510.16053v1>
2. Event-Driven Traffic Management - RIT Digital Institutional Repository, accessed June 16, 2026,

- <https://repository.rit.edu/cgi/viewcontent.cgi?article=13578&context=theses>
3. Astram event data_anonymized - Astram event data_anonymizedb40ac87.csv
 4. Machine Learning Software to Predict Traffic Congestion - dida.do, accessed June 16, 2026, <https://dida.do/use-cases/traffic-congestion-prediction>
 5. An Optimization-Based Traffic Diversion Model during Construction Closures | COSMOS, accessed June 16, 2026, <https://cosmos.uta.edu/wp-content/uploads/2020/04/COSMOS-18-02.pdf>
 6. An Optimization-Based Traffic Diversion Model During Construction Closures, accessed June 16, 2026, <https://workzonesafety.org/publication/an-optimization%E2%80%90based-traffic-diversion-model-during-construction-closures/>
 7. ASTraM: Actionable Intelligence for Sustainable Traffic Management - INSIGHTS IAS, accessed June 16, 2026, <https://www.insightsonindia.com/2026/02/27/astram-actionable-intelligence-for-sustainable-traffic-management/>
 8. Predicting Crash-Related Incident Clearance Time on Louisiana's Rural Interstate Using Ensemble Tree-Based Learning Methods - MDPI, accessed June 16, 2026, <https://www.mdpi.com/2076-3417/14/23/10964>
 9. Traffic Prediction with Machine Learning: How to Forecast Co - AltexSoft, accessed June 16, 2026, <https://www.altexsoft.com/blog/traffic-prediction/>
 10. Forecasting Road Incident Duration Using Machine Learning Framework - SCIRP, accessed June 16, 2026, <https://www.scirp.org/journal/paperinformation?paperid=141218>
 11. [D] Need suggestion for Traffic prediction Model : r/MachineLearning - Reddit, accessed June 16, 2026, https://www.reddit.com/r/MachineLearning/comments/1njbzj8/d_need_suggestion_for_traffic_prediction_model/
 12. Traffic4cast at NeurIPS 2021 – Temporal and Spatial Few-Shot Transfer Learning in Gridded Geo-Spatial Processes, accessed June 16, 2026, <https://proceedings.mlr.press/v176/eichenberger22a/eichenberger22a.pdf>
 13. Finding an Optimal Route: Minimizing Police Stops - Stanford University, accessed June 16, 2026, <https://web.stanford.edu/class/aa228/reports/2019/final107.pdf>
 14. Machine Learning Traffic Flow Prediction Models for Smart and Sustainable Traffic Management - MDPI, accessed June 16, 2026, <https://www.mdpi.com/2412-3811/10/7/155>
 15. The Application Prospects of Event Data in Traffic Flow Prediction - ITM Web of Conferences, accessed June 16, 2026, https://www.itm-conferences.org/articles/itmconf/pdf/2025/01/itmconf_dai2024_01011.pdf
 16. [2209.08735] Traffic incident duration prediction via a deep learning framework for text description encoding - arXiv, accessed June 16, 2026, <https://arxiv.org/abs/2209.08735>
 17. VikasSingh-DS/kaggle-Traffic-Congestion-Prediction - GitHub, accessed June 16, 2026, <https://github.com/VikasSingh-DS/kaggle-Traffic-Congestion-Prediction>
 18. BigQuery-Geotab Intersection Congestion - Kaggle, accessed June 16, 2026,

<https://www.kaggle.com/competitions/bigquery-geotab-intersection-congestion/data>

19. BigQuery-Geotab Intersection Congestion | Kaggle, accessed June 16, 2026, <https://www.kaggle.com/competitions/bigquery-geotab-intersection-congestion>
20. Can you predict wait times at major city intersections - Geotab, accessed June 16, 2026, <https://www.geotab.com/blog/wait-times-major-cities/>
21. Insightful EDA + modeling LGBM hyperopt - Kaggle, accessed June 16, 2026, <https://www.kaggle.com/code/kabure/insightful-eda-modeling-lgbm-hyperopt>
22. Winning Team Submission Traces - Kaggle, accessed June 16, 2026, <https://www.kaggle.com/code/jtrotman/winning-team-submission-traces>
23. Traffic4cast competition reveals novel way to predict traffic flow using AI, accessed June 16, 2026, <https://www.here.com/about/press-releases/2019-14-12>
24. [2203.17070] Traffic4cast at NeurIPS 2021 -- Temporal and Spatial Few-Shot Transfer Learning in Gridded Geo-Spatial Processes - arXiv, accessed June 16, 2026, <https://arxiv.org/abs/2203.17070>
25. Traffic4cast 2022 Competition: from few public vehicle counters to entire city-wide traffic - GitHub, accessed June 16, 2026, <https://github.com/iarai/NeurIPS2022-traffic4cast>
26. Looking 15 minutes into the future | ITS International, accessed June 16, 2026, <https://www.itsinternational.com/its2/its5/its6/its8/news/looking-15-minutes-future>
27. iarai/NeurIPS2021-traffic4cast: Code accompanying our NeurIPS 2021 traffic4cast challenge - GitHub, accessed June 16, 2026, <https://github.com/iarai/NeurIPS2021-traffic4cast>
28. Open Policing Project - Computational Policy Lab, accessed June 16, 2026, <https://policylab.hks.harvard.edu/projects/stanford-open-policing-project.html>
29. samudraneel05/stanford-open-policing: ML to analyze police interactions on traffic stops across the United States - GitHub, accessed June 16, 2026, <https://github.com/samudraneel05/stanford-open-policing>
30. Balancing Classifier Fairness with Public Safety in Traffic Stops CS 229 Final Project - Stanford University, accessed June 16, 2026, <https://cs229.stanford.edu/proj2017/final-reports/5244311.pdf>
31. The freeCodeCamp Podcast, accessed June 16, 2026, <https://podcasts.apple.com/no/podcast/the-freecodecamp-podcast/id1313660749>
32. Unlocking Potential: College Hackathons in Computer Vision - alwaysAI, accessed June 16, 2026, <https://alwaysai.co/blog/college-hackathons-computer-vision-change-world>
33. Can Machine Learning Save Us From Being Stuck in Traffic? - Technology and Operations Management, accessed June 16, 2026, <https://aiinstitute.hbs.edu/platform-rctom/submission/can-machine-learning-save-us-from-being-stuck-in-traffic/>
34. City Hall's Technology Journey - Harvard Kennedy School, accessed June 16, 2026, <https://www.hks.harvard.edu/sites/default/files/centers/rappaport/files/bch%20te>

[chnology%20v4_0.pdf](#)

35. Waze Traffic Tool - Data-Smart City Solutions, accessed June 16, 2026, <https://datasmart.hks.harvard.edu/solutions/waze-traffic-tool>
36. Should Cities Use Hackathons to Solve Social Problems? Lessons from America's Datafest at Harvard, accessed June 16, 2026, <https://studentreview.hks.harvard.edu/should-cities-use-hackathons-to-solve-social-problems-lessons-from-americas-datafest-at-harvard/>
37. Harvard research helps prepare policy road for self-driving cars, accessed June 16, 2026, <https://news.harvard.edu/gazette/story/2019/01/harvard-research-helps-prepare-policy-road-for-self-driving-cars/>
38. AI and Machine Learning in Traffic Flow Prediction: Enhancing Urban Mobility - Eltec Corp, accessed June 16, 2026, <https://elteccorp.com/ai-machine-learning-traffic-flow-prediction-urban-mobility/>
39. Forecasting Road Incident Duration Using Machine Learning Framework - Iowa State University, accessed June 16, 2026, <https://dr.lib.iastate.edu/bitstreams/d2359dc4-bb37-4735-8399-8424536ced63/download>
40. Prediction of Duration of Traffic Incidents by Hybrid Deep Learning Based on Multi-Source Incomplete Data - PMC, accessed June 16, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9518162/>
41. Prediction of Traffic Incident Clearance Duration Using Neural Network for Multimodal Data Distribution | Journal of Transportation Engineering, Part A - ASCE Library, accessed June 16, 2026, <https://ascelibrary.org/doi/10.1061/JTEPBS.TEENG-7923>
42. Traffic Incident Impact Prediction Using Machine Learning and Explainable AI: Evidence from Istanbul - MDPI, accessed June 16, 2026, <https://www.mdpi.com/2079-9292/15/6/1162>
43. Traffic Congestion Prediction Using Machine Learning Techniques - arXiv, accessed June 16, 2026, <https://arxiv.org/html/2206.10983v5>
44. A Systematic Literature Review of Traffic Congestion Forecasting: From Machine Learning Techniques to Large Language Models - MDPI, accessed June 16, 2026, <https://www.mdpi.com/2624-8921/7/4/142>
45. Proposal of a Machine Learning Approach for Traffic Flow Prediction - PMC, accessed June 16, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11014399/>
46. AGENT-BASED MODEL SIMULATION FOR POLICE DEPLOYMENT DECISION-MAKING IN PATROL OPERATIONS - MavMatrix, accessed June 16, 2026, https://mavmatrix.uta.edu/cgi/viewcontent.cgi?article=1185&context=industrialmath_nusys_dissertations
47. Analysis of Traffic Congestion Using LSTM and Graph Theory - SciTePress, accessed June 16, 2026, <https://www.scitepress.org/Papers/2024/128057/128057.pdf>
48. Evaluating Police Department Policy Decisions Using a Simulation Model of Sworn

- Officer Deployment, accessed June 16, 2026,
<https://www.ojp.gov/ncjrs/virtual-library/abstracts/evaluating-police-department-policy-decisions-using-simulation>
49. Artificial Intelligence-based Optimization Models for the Technical Workforce Allocation and Routing Problem Considering Product - The Science and Information (SAI) Organization, accessed June 16, 2026,
https://thesai.org/Downloads/Volume14No12/Paper_33-Artificial_Intelligence_based_Optimization_Models.pdf
 50. The Traffic Police Location and Schedule Assignment Problem | Request PDF - ResearchGate, accessed June 16, 2026,
https://www.researchgate.net/publication/261569659_The_Traffic_Police_Location_and_Schedule_Assignment_Problem
 51. Predictive Policing using Machine Learning (With Examples) - Cogent Infotech, accessed June 16, 2026,
<https://cogentinfo.com/resources/predictive-policing-using-machine-learning-with-examples>
 52. Ramp Management and Control Handbook FINAL - FHWA Operations, accessed June 16, 2026,
https://ops.fhwa.dot.gov/publications/ramp_mgmt_handbook/manual/manual/pdf/rm_handbook.pdf
 53. Executive Summary - NOAA Office for Coastal Management, accessed June 16, 2026, https://coast.noaa.gov/data/hes/docs/general_info/SEUS_HES.pdf
 54. An optimization-based traffic diversion model during construction closures - ResearchGate, accessed June 16, 2026,
https://www.researchgate.net/publication/335401138_An_optimization-based_traffic_diversion_model_during_construction_closures
 55. Bangalore Traffic Haul, using DATA and AI - Reddit, accessed June 16, 2026,
https://www.reddit.com/r/bangalore/comments/1m7zpae/bangalore_traffic_haul_using_data_and_ai/
 56. Advanced Traffic Signal Control Algorithms - Caltrans, accessed June 16, 2026,
<https://dot.ca.gov/-/media/dot-media/programs/research-innovation-system-information/documents/final-reports/ca13-2157a-finalreport-a11y.pdf>
 57. NAVAL POSTGRADUATE SCHOOL THESIS - DTIC, accessed June 16, 2026,
<https://apps.dtic.mil/sti/pdfs/AD1164399.pdf>
 58. REFOL: Resource-Efficient Federated Online Learning for Traffic, accessed June 16, 2026, <https://arxiv.org/html/2411.14046v1>
 59. (PDF) New Perspectives on the Use of Online Learning for, accessed June 16, 2026,
https://www.researchgate.net/publication/347019861_New_Perspectives_on_the_Use_of_Online_Learning_for_Congestion_Level_Prediction_over_Traffic_Data
 60. Optimal Transport based Drift Detection for Sensor Streams: Method and Applications in Transportation - IIM Ahmedabad, accessed June 16, 2026,
<https://www.iima.ac.in/sites/default/files/rnpfiles/16464727852021-09-01.pdf>
 61. New Perspectives on the Use of Online Learning for Congestion Level Prediction over Traffic Data - arXiv, accessed June 16, 2026, <https://arxiv.org/pdf/2003.14304>

62. Transforming Bengaluru traffic with the ASTraM initiative - Arcadis, accessed June 16, 2026,
<https://www.arcadis.com/en/projects/asia/india/transforming-bengaluru-traffic-with-the-astram-initiative/>
63. Bengaluru Traffic Policing Goes High-Tech: 87% Violations Detected by AI Cameras in 2025, accessed June 16, 2026,
<https://www.industrialautomationindia.in/news/bengaluru-ai-cameras-traffic-violations-automation-2025>